

- 1 (a) During translation, a polypeptide is synthesised when amino acids are added to a growing chain of amino acids.

Fig. 1.1 shows part of a growing chain of amino acids and the amino acid cysteine.

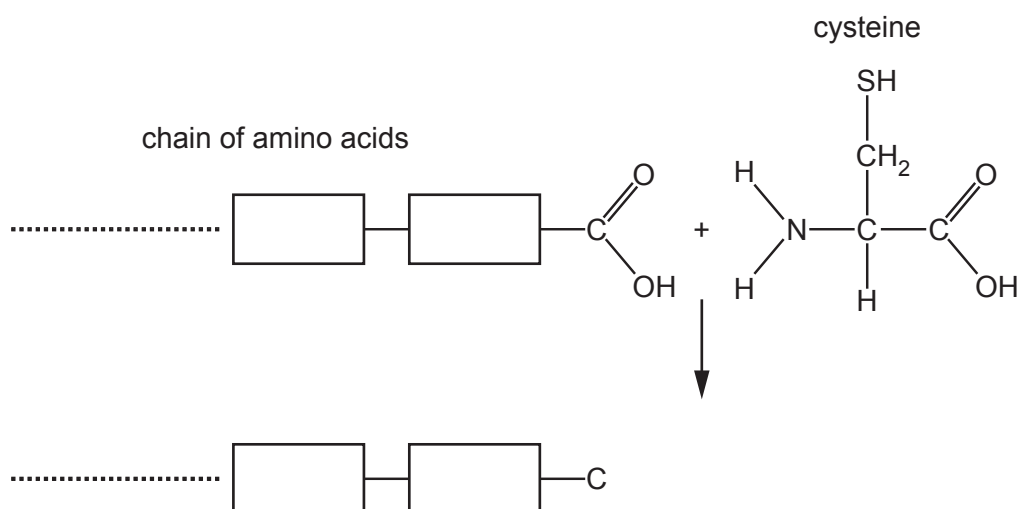


Fig. 1.1

- (i) Complete Fig. 1.1 by showing the formation of the bond between cysteine and the growing chain of amino acids in the process of translation. [3]

- (ii) State the name of the covalent bond that forms when cysteine is added to the growing chain of amino acids.

..... [1]

- (iii) State the organelle where the reaction shown in Fig. 1.1 takes place.

..... [1]

- (b) Fig. 1.2 is a ribbon diagram showing the three-dimensional structure of a protein from the bacterium *Streptococcus*.

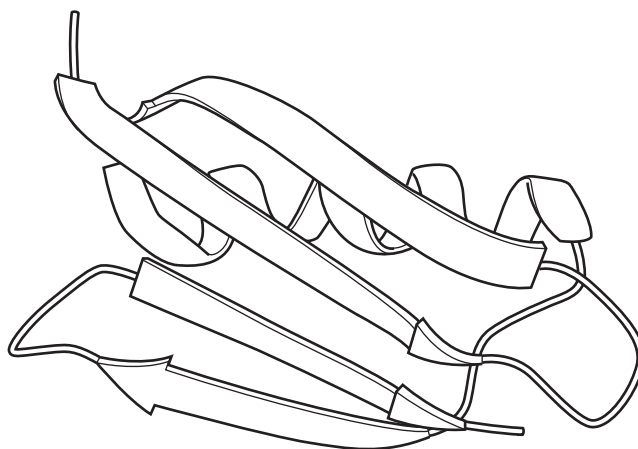


Fig. 1.2

(i) Describe the secondary structure of the protein shown in Fig. 1.2.

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..... [2]

(ii) Explain why the protein shown in Fig. 1.2 has tertiary structure, but not quaternary structure.

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..... [2]

(iii) An analysis of the amino acid composition of the protein in Fig. 1.2 showed that it does **not** contain any cysteine residues.

Explain how the three-dimensional structure of the protein shown in Fig. 1.2 is held in place.

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..... [3]

[Total: 12]